

CSCI-246 Discrete Structures HW 04

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Objective

- Understanding relations and induction principle.
- Mathematical definitions.
- How to approach solving a problem.

Submission requirements

- *Type or clearly hand-write* your solutions into a **PDF FORMAT**.
- **DO NOT UPLOAD images.**
- *non-pdf or emailed solutions will not be graded.*
- If you take pictures of your handwritten homework, put it into pdf format.
- *Start each problem in a new page.*
- Follow the model that you have learned during the lectures for proofs.
- Do not wait until the last minute to submit the assignment.
- You can submit any number of times before the deadline.
- If you are using latex, and you do not know how to type a symbol, use the following website. You can draw the symbol here and it will give you the latex code and the packages that you have to import. <https://detexify.kirelabs.org/classify.html>

- If you are using latex to write the answer, you can use overleaf to make your life easier. **Overleaf is a free, online platform that helps users create and publish scientific and technical documents using LaTeX, a markup-based document preparation system**
- If you do not understand a problem, ask questions during/after the lectures, or during office hours or via discord.
- ***Do not use generative AI to write answers.***

1 Q1

Suppose you are given three relations R_1, R_2, R_3 defined on the set $S = \mathbb{P}(\{0, 1, 2, 3\})$. Note that $R_1, R_2, R_3 \subseteq S \times S$. For each of these relations determine whether:

- it is reflexive, irreflexive, or neither,
 - it is symmetric, anti-symmetric, both, or neither,
 - it is transitive or not, and
 - all pairs of elements are comparable (that is, $\forall a \neq b \in S : (aRb \vee bRa)$).
1. $R_1 \subseteq S \times S$ such that $(A, B) \in R_1$ if (i) A and B are nonempty and the largest element in A equals the largest element in B , or (ii) if $A = B = \emptyset$.
 2. $R_2 \subseteq S \times S$ such that $(A, B) \in R_2$ if the sum of elements in A is equal to the sum of elements in B . (Formally, $A R_2 B \iff \sum_{x \in A} x = \sum_{y \in B} y$.)
 3. $R_3 \subseteq S \times S$ such that $(A, B) \in R_3$ if $A \cap B \neq \emptyset$.

Hint: Write down the set S and $S \times S$. Then try to see which elements in $S \times S$ are related to each other under R_1, R_2, R_3 relations.

2 Q2

In this problem, you will learn some new definitions. Note that we can define the size of a relation to be the number of elements in the corresponding set. For example, consider the set $A = \{1, 2, 3, 4\}$ and the relation $R = \{\langle 2, 4 \rangle, \langle 4, 3 \rangle, \langle 4, 4 \rangle\}$. We say that $|R| = 3$.

We define the closure of a relation R with respect to some property as the smallest relation that includes everything from R but also has the property. To be precise:

- The reflexive closure of a relation R is the smallest possible $R' \supseteq R$ such that R' is reflexive.
- The symmetric closure of a relation R is the smallest possible $R' \supseteq R$ such that R' is symmetric.
- The transitive closure of a relation R is the smallest possible $R' \supseteq R$ such that R' is transitive.

If R already has the property, then the closure is just R . For example, the reflexive closure of the $=$ relation on a set is just the $=$ relation, since $=$ is already reflexive.

For the set A and relation $R \subseteq A \times A$ given above, give the following:

1. The reflexive closure of R .
2. The symmetric closure of R .
3. The transitive closure of R .

3 Q3

List a partial order, strict partial order and a equivalence relation that you can create from the following set $A = \{0, 1, 2, 3\}$.

4 Q4

For a given integer $n \geq 0$, consider the sum of first n cubes: $0^3 + 1^3 + 2^3 + \cdots + n^3$. You can write the sum of first n cubes as $\sum_{i=0}^n i^3$. We can hypothesize that this sum of cubes of first n natural numbers are $\sum_{i=0}^n i^3 = \left(\frac{n \cdot (n+1)}{2}\right)^2$.

1. Show that this formula works for $0 \leq n \leq 3$. (This is very simple try to manually check the sum of cubes for $0 \leq n \leq 3$ and check whether you get the same value from the formula.)

2. Use mathematical induction to prove that this formula works for any integer $n \geq 0$.

Hint: Try to model this problem into the induction framework that we learnt during the class.

- define the predicate $P(n)$.
- State the variable that you are performing the induction over.
- State the base case.
- prove the base case.
- state the inductive case.
- prove the inductive case.
 - assume the inductive hypothesis $P(n - 1)$
 - start with the **LHS** of the $P(n)$ and manipulate it to get the **RHS** (or vice versa.)
Do not start with LHS = RHS.
 - find a way to get the $P(n - 1)$ in your algebra somewhere, so you can apply the inductive hypothesis.
 - correctly apply the inductive hypothesis.
 - clearly say that you have applied the inductive hypothesis.
 - Then derive the **RHS** of the $P(n)$ (if you start with the **LHS**) or **LHS** (if you start with the **RHS**).
- Finish the proof by tying everything together.

5 Q5

Suppose you want to calculate the sum of first n odd numbers. For example, the sum of first 3 odd numbers would be $1 + 3 + 5 = 9$. We can label this sum as $\sum_{i=1}^n (2i - 1)$. We can hypothesize that this sum is equal to n^2 .

1. Show that this formula works for $1 \leq n \leq 3$. (This is simple, manually check whether the **LHS** and **RHS** of the claim is equal).
2. Use mathematical induction to prove that this formula is correct for $n \geq 1$.

Hint: Try to model this problem into the induction framework that we learnt during the class.

- define the predicate $P(n)$.
- State the variable that you are performing the induction over.
- State the base case.

- prove the base case.
- state the inductive case.
- prove the inductive case.
 - assume the inductive hypothesis $P(n - 1)$
 - start with the **LHS** of the $P(n)$ and manipulate it to get the **RHS** (or vice versa.)
Do not start with $\text{LHS} = \text{RHS}$.
 - find a way to get the $P(n - 1)$ in your algebra somewhere, so you can apply the inductive hypothesis.
 - correctly apply the inductive hypothesis.
 - clearly say that you have applied the inductive hypothesis.
 - Then derive the **RHS** of the $P(n)$ (if you start with the **LHS**) or **LHS** (if you start with the **RHS**).
- Finish the proof by tying everything together.